

SULAMAN KHAN

Protocol Engineer | Rust • Go • EVM Internals • Layer 1 Infrastructure

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Full overlap with Gulf hours • 3–4 hours with UK/EU • US mornings covered

PROFESSIONAL SUMMARY

Systems engineer who shipped a live Layer 1 chain with an EVM compatibility layer, and the full product surface around it — block explorer, Web3 wallet, DEX, NFT marketplace and bridge. My protocol work covers consensus configuration, validator-set and node-authorization pallets, node and networking behaviour, load testing and gas-limit tuning; I have also implemented proof-of-stake consensus from scratch in Go. Rust and Go are both production languages for me, alongside eight years of backend engineering: an API tuned to 1M+ requests, a 1M-user service migrated from Node.js to Go, and a fintech platform that moved \$1M+ in its first quarter. Public chains are an unusually strict production environment, and that is the discipline I bring.

CORE SKILLS

Protocol & consensus: Substrate / Polkadot SDK, FRAME pallet development, Frontier (EVM compatibility layer), Aura consensus, proof-of-stake consensus implementation, validator-set and node-authorization pallets, node synchronization, block production

Languages: Rust (async/tokio, Actix), Go, Solidity, Python, Node.js, TypeScript

Chain engineering: EVM internals, JSON-RPC, transaction signing, DeFi protocol integration, smart-contract interaction, blockchain load testing, gas-limit tuning, bulk-transaction throughput analysis

Systems & infra: gRPC, Protocol Buffers, P2P and networking protocols, microservices, concurrency (goroutines, channels), PostgreSQL, Redis, Elasticsearch, GraphQL

Cloud & DevOps: AWS, Docker, Kubernetes, Nginx, DigitalOcean, GitHub Actions, CI/CD, Datadog

AI / LLM: Python, FastAPI, LangChain, LangGraph, RAG pipelines, LLM agents, Model Context Protocol (MCP) — relevant to AI-adjacent protocol work

PROFESSIONAL EXPERIENCE

Blockchain & Backend Engineer → AI Engineer

Dec 2021 – Present

Vaival Technologies • Client: Vulcan Forged (US) • Remote

- **Layer 1 core engineering:** Core engineer on Elysium Chain, a live Substrate-based Layer 1 with a Frontier EVM compatibility layer — Aura consensus configuration, validator-set and node-authorization pallets, and node behaviour under load.
- **Chain performance:** Ran blockchain load testing covering bulk transaction throughput and gas-limit tuning; optimized a chain-adjacent API to handle 1M+ requests through server-resource management.
- **Protocol product surface:** Built the block explorer, Web3 CloudWallet, DEX, NFT marketplace, and bridge and swap modules in Go and Node.js, serving real-time on-chain data over GraphQL.
- **Security-critical paths:** Implemented transaction signing, DeFi protocol integration and secure smart-contract interaction on a public network, where state is permanent and defects are visible to everyone.
- **Systems performance:** Migrated a Node.js service serving 1M users to Go, resolving persistent memory issues; raised throughput with goroutines and channels and gRPC microservices.
- **AI systems (Jan 2024–present):** Built production RAG pipelines, LLM agents and MCP server/client integrations as async FastAPI services with tenant isolation, cost and latency observability, and multi-provider failover.

Full Stack Engineer (Web & Mobile)

Jun 2018 – Dec 2021

B4USoft Pvt. Ltd. • Lahore, Pakistan

- **Fintech:** Built secure transaction, authentication and fraud-prevention features; the launched application processed \$1M+ in transactions in its first quarter.
- **Scale & cost:** Optimized a CMS to 100,000+ daily active users; moved workloads to serverless on AWS Lambda, cutting operational costs 40%. Led projects to CMMI 2.0 certification.

SELECTED PROJECTS

Note on code visibility: production work for client engagements lives in client-owned repositories. Public architecture write-ups for the systems below are published at github.com/sulaman0.

- **Elysium Chain** — Live Layer 1 (Substrate + EVM) for gaming, NFT and metaverse projects, with Elysium Scan block explorer, CloudWallet Web3 wallet, and an AI NFT studio. elysiumchain.tech
- **VulcanX Exchange** — Gamified exchange with revenue sharing, weekly buybacks and leaderboards. Apr-Sep 2025.
- **Substrate Aura Core Chain** — Substrate chain built with Aura consensus, integrating the Substrate validator-set pallet and node authorization. Rust.
- **PoS Consensus in Go** — Proof-of-stake consensus from scratch: multiple validator nodes, stake assignment, stake-weighted random leader selection, block proposal and cross-node validation, chain state append.
- **Blockchain Load Testing** — Harness for chain load, bulk transaction throughput and gas-limit behaviour under stress.
- **Actix Rust** — Actix-web service implementation in Rust, async/tokio patterns.
- **ZATCA Phase 2 E-Invoicing** — Saudi tax-compliance implementation with XML generation and certificate signing. Open-sourced; 13 stars, 7 forks.

SELECTED ACHIEVEMENTS

- Core engineer on a live Layer 1 chain with an EVM compatibility layer, plus its explorer, wallet, DEX, marketplace and bridge.
- Optimized a chain-adjacent API to handle 1M+ requests; migrated a 1M-user service from Node.js to Go.
- Fintech application processed \$1M+ in transactions within its first quarter post-launch.
- Cut operational costs 40% through serverless architecture; led projects to CMMI 2.0 certification.

EDUCATION

B.S. Information Technology

2014 - 2018

University of the Punjab, Lahore, Pakistan

- Applied Concurrency in Go — goroutines, channels, concurrent systems design (2023)